USN [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] [22TCSE216]

FIFTH SEMESTER B.Tech EXAMINATION, NOVEMBER-2024

CLOUD COMPUTING AND VIRTUALIZATION

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A	**5 x2=10**		
1.	**a.**	Explain virtualization with its benefits.	L2	1	**02**
	b.	Illustrate the key concept in cloud deployment.	L2	2	**02**
	c.	Explain the cloud object storage with example.	L2	3	**02**
	d.	Illustrate different types of cloud platforms with example.	L2	4	**02**
	e.	Explain the features of VM in the cloud.	L2	5	**02**
		PART-B	**6 X 5=30**		
2.	**a.**	Explain different types of cloud services with example	L2	CO1	**05**
		OR			
	b.	Identify the key functionalities and explain Infrastructure-as-a-Service (IaaS) with example.	L2	CO1	**05**
3.	**a.**	Illustrate the process of cloud deployment and management in e-commerce application	L2	CO2	**05**
		OR			
	b.	Compare different deployment models with examples	L4	CO2	**05**
4.	**a.**	Explain AWS Elastic Beanstalk and discuss its features, benefits, and its use cases.	L4	CO3	**05**
		OR			
	b.	Explain data security in cloud with examples	L3	CO3	**05**
5.	**a.**	Explain the cloud block storage with example	L3	CO4	**05**
		OR			
	b.	Compare and contrast the concept of cloud load balancing and Scaling with example.	L4	CO4	**05**
6.	**a.**	Examine the advantages and challenges in cloud platform	L4	CO5	**05**
		OR			
	b.	Write short notes on AWS Networking and Content Delivery.	L4	CO5	**05**
7.	**a.**	Identify the Common Use Cases for VMs in the Cloud and elaborate with example.	L3	CO6	**05**

		OR			
	b.	Identify the Key Features of Kubernetes.	L3	CO6	**05**
		PART-C	**6 X 10=60**		
8.	**a.**	Illustrate how a Type-1 hypervisor operates, including its architecture, key features, and provide relevant examples with a well-labeled diagram.	L2	CO1	**10**
		OR			
	b.	Compare the features of VM ware, Hyper-V and KVM in detail.	L4	CO1	**10**
9.	**a.**	Explain different steps followed in managing the cloud resources.	L2	CO2	**10**
		OR			
	b.	Analyze a case study demonstrating the strategies implemented to protect sensitive data in healthcare. Evaluate the effectiveness of these strategies.	L5	CO2	**10**
10.	**a.**	Compare the features of VPC and VPN with example.	L4	CO3	**10**
		OR			
	b.	Analyze the case study of AWS S3 in Netflix application with example focusing on how it solves scalability and performance challenges.	L4	CO3	**10**
11.	**a.**	Explain AWS Management Governance and Analytics service in detail with examples.	L2	CO4	**10**
		OR			
	b.	Explain GCP compute and storage services in detail, providing specific examples.	L2	CO4	**10**
12.	**a.**	Examine the key concept of Virtual Desktop Infrastructure (VDI) in the cloud in detail.	L4	CO5	**10**
		OR			
	b.	Identify the main differences between Kubernetes Pods and Docker containers.	L3	CO5	**10**
13.	**a.**	Analyze how does Kubernetes handle scaling and auto-scaling.	L4	CO6	**10**
		OR			
	b.	Compare on GCP networking , big data & machine learning services in detail with examples.	L4	CO5	**10**